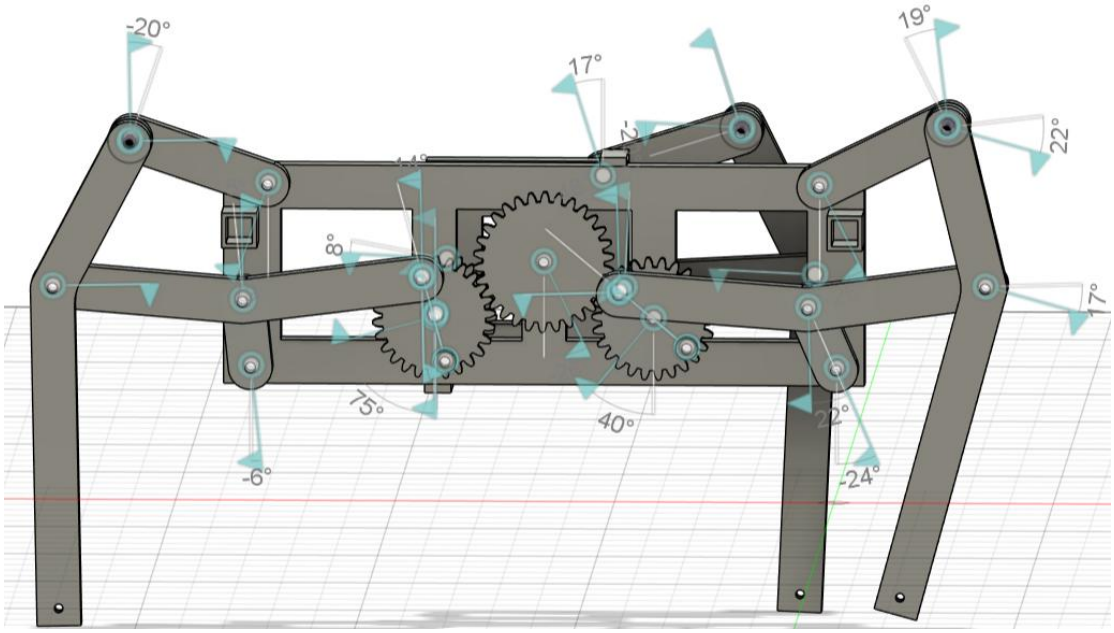
Mechatronic Design Individual Lab Report (ILR04)
Name: Hongfei
Team: Team 2
Teammates: Hongfei Liu, Hongru Liu, Daniel Lazaro, Yechen Zheng
ILR Number: ILR04
Submission Date: Feb. 26, 2026

Individual Progress

Since the last lab demo, I designed and fabricated a functional prototype of one leg module for our gear-driven hexapod subsystem. The primary objective was to validate whether a single main driving gear could actuate three Klann-based legs simultaneously. This concept aims to reduce the total number of motors required in the final robot while maintaining coordinated leg motion.



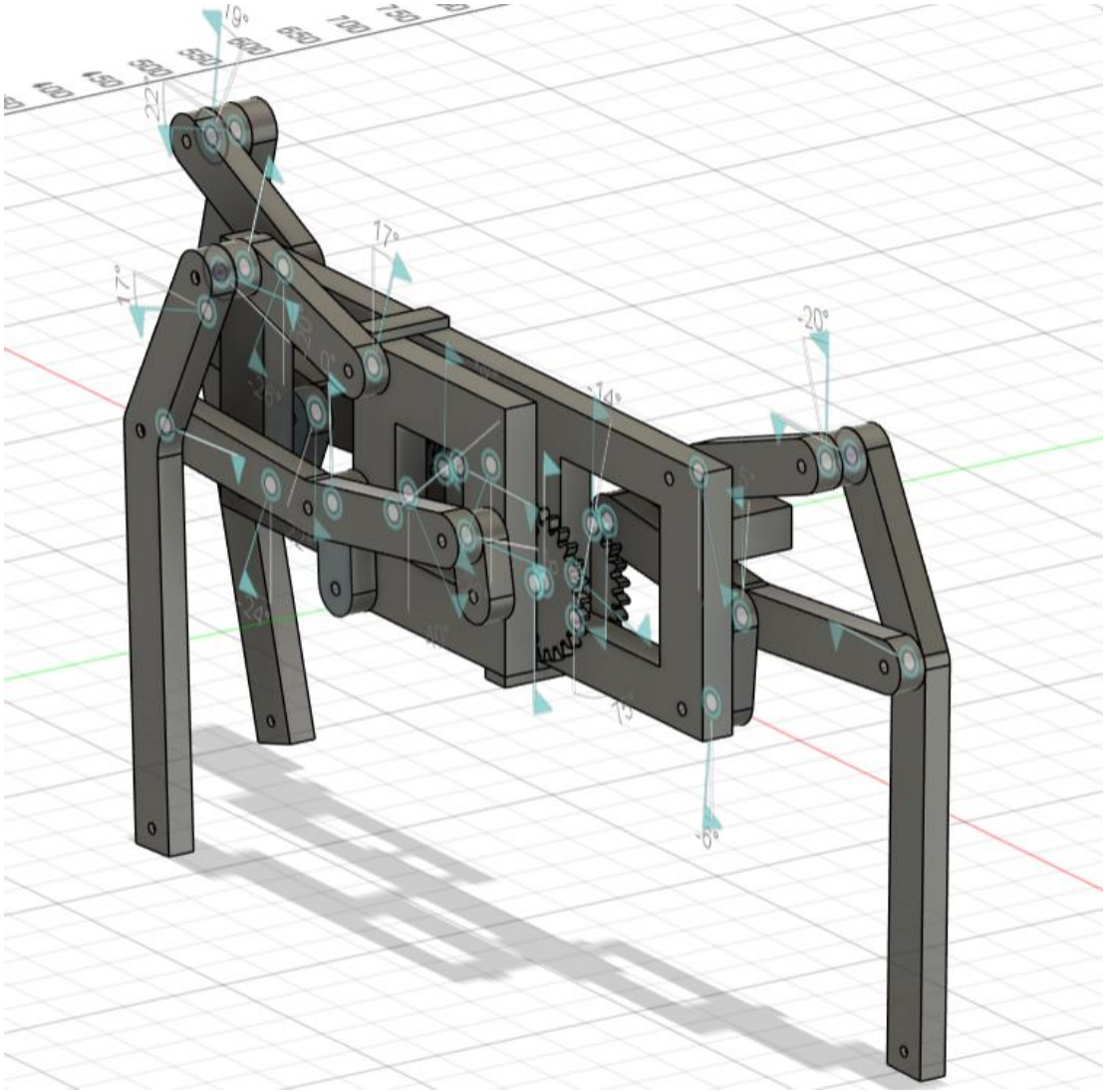


Figure 1: The shape of the prototype. Please forgive me for forgetting to take a picture of the prototype.

The prototype was constructed using laser-cut wooden plates. All structural members and linkages were modeled in CAD before fabrication to approximate the intended geometry and gear alignment. After laser cutting, I manually assembled the linkage system and integrated the central driving gear with three leg mechanisms spaced evenly around the gear axis.

The prototype successfully demonstrates that one main gear can transmit motion to three legs through the linkage connections. When manually rotating the main gear, all three legs move in a synchronized cyclic pattern. This confirms the feasibility of the mechanical transmission concept and validates our high-level design approach of motor reduction through mechanical coupling.

Challenges

Two primary challenges emerged during testing.

First, several components are not fully constrained or securely fixed. While the system operates under manual rotation, some linkage joints and structural elements loosen over time. In certain cases, parts partially detach due to vibration and dynamic loading. This issue likely results from insufficient fastening mechanisms and the absence of rigid joint interfaces. The current prototype relies primarily on friction-fit joints and simple press connections in wood, which do not provide adequate retention

To address this issue, we must redesign connection interfaces in the CAD model. Potential solutions include adding bolt-through joints, shaft collars, spacers to control axial movement, and mechanical locking features such as tabs or interlocking slots. A more robust fastening strategy will be required in the final aluminum or 3D-printed implementation.

Second, tolerance in the current prototype is relatively large. Laser-cut wood introduces dimensional variability due to material thickness inconsistency and kerf width. As a result, several joints exhibit excessive clearance, producing backlash in the linkage motion. The gear meshing is also not optimally tight, leading to slight misalignment during rotation. This reduces motion smoothness and may increase wear if replicated in the final design.

The observed tolerance stack-up suggests that our CAD design needs tighter dimensional control and possibly revised shaft alignment features. Incorporating bushings or bearings in the final design may improve rotational stability and reduce friction-induced misalignment.

Teamwork

This week, our work progressed across mechanical, electrical, and software subsystems.

I refined the leg design and built the wooden prototype using spare materials, validating the single-gear, three-leg concept. Daniel refined the power schematic, ensuring that the electrical architecture aligns with updated motor load expectations. Hongru finalized the motor specifications based on torque and speed requirements.

Yecheng and I ordered initial parts and finalized materials for the frame and chassis to ensure structural compatibility with the leg subsystem. Meanwhile, Daniel and Yecheng demonstrated the ROS behavior tree, establishing the control framework that will eventually coordinate the mechanically coupled legs.

Plans

For this prototype, I will modify the leg assembly and gear mounting interfaces to include more secure fastening mechanisms. This includes adding bolt holes, shaft retention features, and improved structural supports, reduce tolerance and improve alignment.

I will also focus on improving terrain adaptability, drivetrain validation, and geometric precision. A key priority is designing a passive compliance mechanism for each leg. The goal of this system is to absorb shock loads and accommodate uneven terrain without transmitting excessive force into the drivetrain. I plan to investigate spring-based solutions, potentially incorporating compression or torsional springs positioned strategically within the linkage or foot assembly. The spring stiffness will be estimated using static load assumptions to ensure sufficient support while allowing limited vertical compliance.